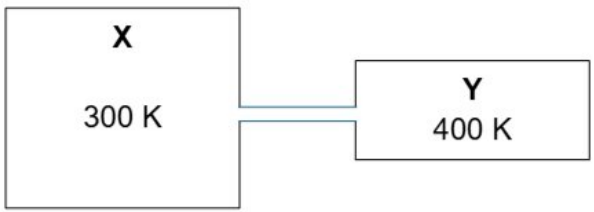
17 Two containers X and Y are connected by a narrow tube as shown below.



The volume of container X is twice that of container Y.

Both containers are filled with an ideal gas and the steady state is achieved with X and Y held at 300 K and 400 K respectively.

If the mass of gas in X is m, what is the mass of gas in Y?

A $\frac{1}{4}m$ **B** $\frac{3}{8}m$ **C** $\frac{3}{4}m$ **D** m